CAMBRIDGE INSTITUTE OF TECHNOLOGY

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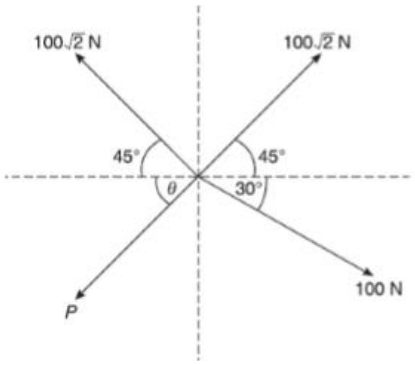
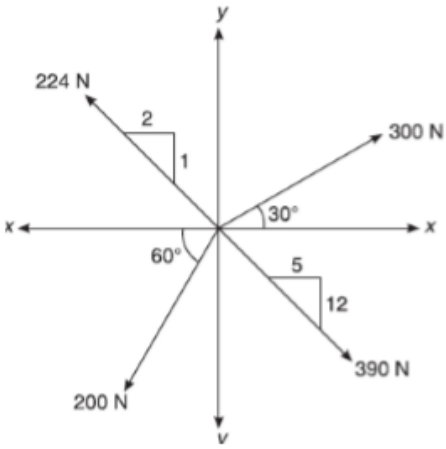
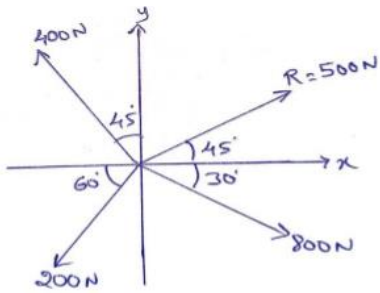
Department of Civil Engineering

ENGINEERING MECHANICS **ASSIGNMENT – 1** **DATE: 18/03/2026**

1. Define Engineering Mechanics. Explain briefly classification of Engineering Mechanics.
2. Define Force and Mention its characteristics.
3. Define i. particle ii. rigid body iii. continuum iv. Deformable body.
4. Define Force system. Explain briefly the classification of force system.
5. State
   i. Principle of Transmissibility of Forces
   ii. Principle of Superposition of Forces
6. What is resolution of force? Explain the method of resolution of force.
7. State and explain Parallelogram Law of forces.
8. Determine the magnitude and direction of force P, which keeps the concurrent system of Figure in equilibrium.



9. Determine the magnitude and direction of the resultant of the coplanar concurrent force system shown in below



10. Determine the fourth unknown force in magnitude and direction so that the resultant R acts as shown in below Fig.